

Tag Corner Detection

Approach Overview

Overview

The task is to detect the four corner coordinates (x_1 y_1 x_2 y_2 x_3 y_3 x_4 y_4) of a fiducial tag in each image. Three distinct approaches were explored: a classical computer vision pipeline, a direct regression deep learning model, and a heatmap-based deep learning model. Each approach also handles the case where no tag is present in the image.

Approach 1 — Classical Computer Vision

Pipeline

This approach uses traditional OpenCV operations without any learned model. The pipeline proceeds through the following stages:

- **Preprocessing:** The image is converted to LAB colour space and CLAHE (Contrast Limited Adaptive Histogram Equalisation) is applied to the L channel to improve contrast uniformity across varying lighting conditions.
- **Binarisation:** The enhanced grayscale image is thresholded using Otsu's method, which automatically selects an optimal threshold to separate foreground from background.
- **Contour Detection:** OpenCV's `findContours` is run on the binary image using `RETR_CCOMP` to retrieve both outer and inner contours.
- **Quadrilateral Filtering:** Each contour is approximated using `approxPolyDP`. Only four-sided, convex contours whose area falls within a defined range (25,000–100,000 px^2) and that do not touch the image border are kept.
- **Corner Sorting:** The four vertices of the best candidate quad are sorted geometrically into Bottom-Right, Bottom-Left, Top-Left, and Top-Right order.

No-Tag Handling

If no qualifying quadrilateral is found after filtering, the image is flagged as `no_quad` and no coordinates are returned. The area bounds and border-margin exclusion act as natural rejection criteria for false positives.

Evaluation

Per-corner pixel error is computed against ground truth. A corner is considered correctly detected if its Euclidean distance from the ground truth is within 10 pixels. Accuracy is reported as the percentage of the four corners matched within this threshold.

Approach 2 — ResNet Direct Regression

Architecture

This is a supervised deep learning approach that frames corner detection as a direct coordinate regression problem. Three backbone variants were explored:

- ResNet-18 with the final fully-connected layer replaced by a linear layer outputting 9 values (8 normalised coordinates + 1 confidence logit). Only the last residual block (layer4) and the new head are trainable; earlier layers are frozen.
- EfficientNet-B0 with its classifier head replaced by the same 9-output linear layer, fully fine-tuned.
- ConvNeXt-Tiny with its classifier head similarly replaced, with the backbone frozen and only the head trained.

Training

Images are resized to 224×224 and normalised with ImageNet statistics. Corner coordinates are normalised by image width and height to fall in [0, 1]. The model is trained with a combined loss: MSE on the 8 coordinate outputs and Binary Cross-Entropy (with logits) on the confidence output. Adam is used as the optimiser with a learning rate of 1e-3 and weight decay of 1e-4.

No-Tag Handling

The model outputs a confidence logit alongside the coordinates. During inference, if the sigmoid of this logit falls below a chosen threshold, the prediction is rejected and no coordinates are returned, handling images where no tag is present.

Approach 3 — Heatmap-Based Detection

Architecture

Instead of regressing coordinates directly, this approach trains the model to produce a spatial heatmap for each of the four corners — essentially a 2D Gaussian centred at each corner location. The model follows an encoder-decoder structure:

- Encoder: A pretrained ResNet-18 backbone (all convolutional layers, excluding the global pool and FC) extracts feature maps down to a 7×7 spatial resolution.
- Decoder: Three transposed-convolution blocks progressively upsample the features back to 56×56, producing four output channels — one heatmap per corner. Each block uses BatchNorm and ReLU.
- Confidence Head: A global average pool on the encoder features feeds a single linear layer to produce a confidence logit, indicating whether a tag is present.

Ground Truth Generation

For each training image, a 56×56 Gaussian heatmap is generated per corner. The Gaussian is centred at the corner's projected position on the 56×56 grid (scaled from original image coordinates) with a sigma of 2 pixels.

Training

The loss is a sum of MSE between predicted and ground-truth heatmaps and BCE on the confidence logit. Corner coordinates are recovered at inference time by taking the argmax of each predicted heatmap and rescaling back to image coordinates.

No-Tag Handling

The confidence head provides an explicit tag-presence score. Predictions are suppressed when the confidence sigmoid is below a set threshold, ensuring no coordinates are returned for images without a tag.

Summary Comparison

Classical CV requires no training data and runs fast, but relies on the tag being well-contrasted and quadrilateral-shaped. It is brittle under complex backgrounds or occlusion.

ResNet Regression is simple and fast at inference, producing coordinates in a single forward pass. It can struggle with spatial precision for small or occluded tags but is easy to train and deploy.

Heatmap Detection retains spatial information through the decoder, making it more precise for corner localisation. It is more complex to train but generally achieves better accuracy, especially when corners are spatially separated.